# AGILE LAB ASSESSMENT – 5

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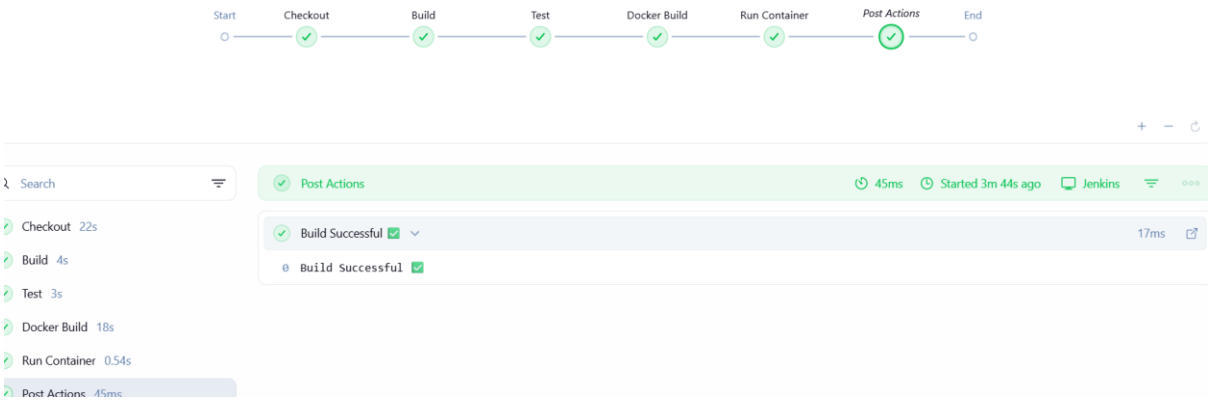
### 22) Contact management System

```
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running ContactServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.044 s - in ContactServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.248 s
[INFO] Finished at: 2026-04-08T17:57:10+05:30
[INFO] ------------------------------------------------------------------------
```

```
[INFO] Running ContactServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.045 s - in ContactServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO]
[INFO] --- jar:3.5.0:jar (default-jar) @ contact-system ---
[INFO] Building jar: C:\Users\rishv\OneDrive\Desktop\contact-system\target\contact-system-1.0.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.534 s
[INFO] Finished at: 2026-04-08T17:57:22+05:30
[INFO] ------------------------------------------------------------------------
```

```
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git init
Reinitialized existing Git repository in C:/Users/rishv/OneDrive/Desktop/contact-system/.git/
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git add .
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git commit -m "first commit"
On branch main
PS C:\Users\rishv\OneDrive\Desktop\contact-system>
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git branch -M main
PS C:\Users\rishv\OneDrive\Desktop\contact-system>
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git remote add origin https://github.com/RishvanthBK/Contact-management-system.git
error: remote origin already exists.
PS C:\Users\rishv\OneDrive\Desktop\contact-system> git push -u origin main
Enumerating objects: 38, done.
Counting objects: 100% (38/38), done.
Delta compression using up to 16 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (38/38), 10.69 KiB | 1.78 MiB/s, done.
Total 38 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/RishvanthBK/Contact-management-system.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\rishv\OneDrive\Desktop\contact-system>
```




```
C:\Users\rishv\.jenkins\workspace\Contact-management-system>docker rm -f contact-container   || exit 0
contact-container

C:\Users\rishv\.jenkins\workspace\Contact-management-system>docker run -d --name contact-container contact-system:latest
31937fc0c66a84f30d11e5cf2b97a75503de1c263991a3e5598d2c218d989de5
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Build Successful ✅
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

```
PS C:\Users\rishv\OneDrive\Desktop\contact-system> kubectl get svc
NAME                    TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
java-service            NodePort    10.96.90.75     <none>        80:30008/TCP   10h
java-test-app-service   NodePort    10.96.174.144   <none>        80:30007/TCP   19h
kubernetes              ClusterIP   10.96.0.1       <none>        443/TCP        8d
PS C:\Users\rishv\OneDrive\Desktop\contact-system>
```

## CODE:

**App.java**

```
public class App {
  public static void main(String[] args) {

    ContactService service = new ContactService();

    service.addContact(new Contact(1, "John", "12345"));

    Contact c = service.getContact(1);
    System.out.println(c.getName());

    service.updateContact(1, "Johnny", "99999");

    System.out.println(service.getContact(1).getName());

    service.deleteContact(1);

    System.out.println("Total contacts: " + service.size());
```

```
    }
}
```

**Contact.java**

```
public class Contact {
    private int id;
    private String name;
    private String phone;

    public Contact(int id, String name, String phone) {
        this.id = id;
        this.name = name;
        this.phone = phone;
    }

    public int getId() { return id; }
    public String getName() { return name; }
    public String getPhone() { return phone; }

    public void setName(String name) { this.name = name; }
    public void setPhone(String phone) { this.phone = phone; }
}
```

**ContactService.java**

```
import java.util.*;

public class ContactService {

  private Map<Integer, Contact> contacts = new HashMap<>();

  // CREATE
  public void addContact(Contact contact) {
    contacts.put(contact.getId(), contact);
  }

  // READ
  public Contact getContact(int id) {
    return contacts.get(id);
  }

  // UPDATE
  public void updateContact(int id, String name, String phone) {
    Contact c = contacts.get(id);
    if (c != null) {
      c.setName(name);
      c.setPhone(phone);
    }
  }

  // DELETE
```

```
    public void deleteContact(int id) {
        contacts.remove(id);
    }

    public int size() {
        return contacts.size();
    }
}
```

**ContactServiceTest.java**

```
import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.*;

public class ContactServiceTest {

    @Test
    void testCRUD() {
        ContactService service = new ContactService();

        // Create
        service.addContact(new Contact(1, "John", "123"));
        assertEquals(1, service.size());

        // Read
        assertEquals("John", service.getContact(1).getName());

        // Update
```

```
        service.updateContact(1, "Doe", "999");
        assertEquals("Doe", service.getContact(1).getName());

        // Delete
        service.deleteContact(1);
        assertEquals(0, service.size());
    }
}
```

**Dockerfile**

```
FROM eclipse-temurin:17-jdk-alpine

WORKDIR /app

COPY target/contact-system-1.0.jar app.jar

CMD ["java", "-jar", "app.jar"]
```

**pom.xml**

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
         http://maven.apache.org/xsd/maven-4.0.0.xsd">

    <modelVersion>4.0.0</modelVersion>
```

```
    <groupId>com.example</groupId>
    <artifactId>contact-system</artifactId>
    <version>1.0</version>

    <dependencies>
      <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter</artifactId>
        <version>5.10.0</version>
        <scope>test</scope>
      </dependency>
    </dependencies>

    <build>
      <plugins>
        <plugin>
          <groupId>org.apache.maven.plugins</groupId>
          <artifactId>maven-surefire-plugin</artifactId>
          <version>3.0.0</version>
        </plugin>
      </plugins>
    </build>

</project>
```

**Jenkins**

```
pipeline {
  agent any

  stages {

    stage('Checkout') {
      steps {
        git branch: 'main', url: 'https://github.com/RishvanthBK/Contact-management-system.git'
      }
    }

    stage('Build') {
      steps {
        bat 'mvn clean package'
      }
    }

    stage('Test') {
      steps {
        bat 'mvn test'
      }
    }

    stage('Docker Build') {
      steps {
```

```
            bat 'docker build -t contact-system:latest .'
        }
    }

    stage('Run Container') {
        steps {
            bat """
            docker rm -f contact-container || exit 0
            docker run -d --name contact-container contact-system:latest
            """
        }
    }

    stage('Kubernetes Deploy') {
        steps {
            bat """
            kubectl apply -f deployment.yaml
            kubectl apply -f service.yaml
            kubectl get pods
            kubectl get services
            """
        }
    }
}

post {
    success {
        echo 'Build + Kubernetes Deploy Successful '
```

```
        }
        failure {
            echo 'Build Failed '
        }
    }
}
```

**deployment.yaml**

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: contact-system
spec:
  replicas: 1
  selector:
    matchLabels:
      app: contact-system
  template:
    metadata:
      labels:
        app: contact-system
    spec:
      containers:
      - name: contact-system
        image: contact-system:latest
        ports:
```

```
      - containerPort: 8080
```

**service.yaml**

```yaml
apiVersion: v1
kind: Service
metadata:
  name: contact-service
spec:
  type: NodePort
  selector:
    app: contact-system
  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30007
```